Code-Layout, Readability and Reusability

Date	29 June 2026
Team ID	xxxxxx
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the code developed for the Credit Card Approval Prediction System. The project follows a well-structured coding approach with meaningful variable names, proper file organization, reusable modules, and clear comments. These practices improve code readability, simplify maintenance, and support future enhancements such as integrating advanced Machine Learning models, database connectivity, or additional prediction features.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing and indentation used throughout the code	Yes	Proper indentation maintained.
2	Proper File Structure	Files and folders are logically organized	Yes	Project organized into src, templates, static, models.
3	Meaningful Variable Names	Variables clearly describe their purpose	Yes	Descriptive and readable variable names used.
4	Function / Method Names	Functions are descriptively named	Yes	Functions clearly represent their functionality.
5	Code Comments	Inline and block comments explain logic	Yes	Comments added to major sections of the code.
6	Modular Design	Code is divided into reusable modules	Yes	Data preprocessing, feature encoding, model training.
7	No Redundant Code	Duplicate or unnecessary code removed	Yes	Clean and optimized implementation.
8	Error Handling	Exceptions and invalid inputs are handled gracefully	Partial	Basic exception handling is implemented in the Flask application.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technolog	Where Reused	Reusability Level (High / Medium / Low)
1	Flask Application	Python, Flask	Handles routing, user input, and prediction	High
2	Data Processing Modules	Python, Pandas	Data analysis, preprocessing, feature encoding, and model training	High
3	HTML Templates	HTML/CSS	Home, Prediction Form, and Result pages	High
4	Trained Machine Learning Model (model.pkl)	Scikit-learn, Joblib	Loaded by the Flask application for prediction	High
5	Label Encoders (label_encoders.pkl)	Scikit-learn, Joblib	Encodes categorical features before prediction	High
6	CSS Stylesheet (style.css)	CSS	Shared across all web pages	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure	5	Well-organized project structure with separate folders for source code, templates, static files, datasets, and models.
Readability	5	Meaningful variable names, proper indentation, and clear code organization improve readability.
Reusability	5	Reusable Flask routes, trained model, label encoders, and HTML templates simplify maintenance and future enhancements.
Documentation / Comments	4	Sufficient comments and documentation explain the Machine Learning workflow and application logic.
Overall Score	5	The project follows good coding standards with a modular, readable, maintainable, and reusable implementation.